

FacilityOS — Woxsen University Executive Overview

Smart Campus Facility Operations, Maintenance Architecture & SLA Engine

Platform Name: FacilityOS Woxsen Core	Version: 2.0 Production
Production Link: facilityos-brown.vercel.app	GitHub Repo: Invarsoft/facilityos-pro
Target Facility: Woxsen University Campus	Deployment: Vercel Edge / Docker

1. Executive Summary

FacilityOS is a custom-engineered campus facility operations platform built specifically for Woxsen University. It digitizes and automates maintenance workflows across Hostel A (Boys), Hostel B (Girls), Academic Blocks 1 & 2, Science & Tech Labs, Central Library, and Campus Amenities. By uniting 3-step student request wizards, trade-accurate technician dispatch, mobile PWA capabilities, and mandatory 4-digit OTP resolution verification, FacilityOS eliminates SLA breaches and manual Warden supervision overhead.

2. System Features & Module Architecture

Module / Feature	Functional Capability	Woxsen Campus Impact
Woxsen Infrastructure Scope	Full coverage for Hostels A/B, Academic Blocks, Mess & Computing Labs.	100% campus data isolation & student portal access.
14 Trade Service Matrix	Plumbing, Electrical, HVAC, IT Networks, Furniture, Cleaning, Elevators, etc.	Tailored service categorization per hostel block.
Specialist Match Engine	Ranks campus technicians by trade skill, active workload, and rating.	Eliminates misassigned tasks & reduces downtime.
Student OTP Sign-Off	4-Digit Security PIN requirement before work order closure.	Prevents unverified or fake ticket mark-as-done.
Categorized PWA Mobile App	Responsive bottom bar navigation with segmented tab filtering for phones.	Instant mobile access for students & ground staff.
5-Level SLA Engine	Real-time countdown timers for Emergency (2h), Critical (4h), High (12h), Normal (24h).	Auto-escalations to Warden & Facility Admin.

3. Technical Architecture & Deployment Stack

- **Frontend:** Next.js 15 (App Router), React 19, TypeScript, Tailwind CSS
- **State Management:** React Context API with Woxsen Campus State
- **Mobile Native:** Progressive Web App (PWA) with CapacitorJS (.ipa / .apk) Support
- **Cloud & Hosting:** Vercel Production Infrastructure & Docker Containerization
- **Security:** Role-Based Access Control (RBAC) & Security Audit Logging

